

Section: Our Dynamic Universe

Topic: Collisions, Explosions and Impulse

Two identical spacecraft, X and Y, are travelling in space. They collide and stick together.

Before the collision:

- X has mass $2.0 \times 10^4 \text{ kg}$ and velocity 6.0 m s^{-1}
- Y has mass $2.0 \times 10^4 \text{ kg}$ and velocity -3.0 m s^{-1}

After the collision, the spacecraft stick together and move as one.

(a)
Calculate the velocity of the spacecraft after the collision

Apply conservation of momentum:

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

Substitute values:

$$(2.0 \times 10^4)(6.0) + (2.0 \times 10^4)(-3.0) = (4.0 \times 10^4)v$$

$$(1.2 \times 10^5) - (6.0 \times 10^4) = 4.0 \times 10^4v \Rightarrow 6.0 \times 10^4 = 4.0 \times 10^4v$$

$$v = \frac{6.0 \times 10^4}{4.0 \times 10^4} = 1.5 \text{ m s}^{-1}$$

(b)
Calculate the total kinetic energy before and after the collision

Before the collision:

$$E_k = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}(2.0 \times 10^4)(6^2) + \frac{1}{2}(2.0 \times 10^4)(3^2)$$

$$= (1.0 \times 10^4)(36) + (1.0 \times 10^4)(9) = 3.6 \times 10^5 + 9.0 \times 10^4 = 4.5 \times 10^5 \text{ J}$$

After the collision:

$$E_k = \frac{1}{2}(4.0 \times 10^4)(1.5)^2 = \frac{1}{2}(4.0 \times 10^4)(2.25) = (2.0 \times 10^4)(2.25) = 4.5 \times 10^4 \text{ J}$$

(c)
Explain why kinetic energy is not conserved in this collision

✔ The spacecraft **stick together**, so this is an **inelastic collision**.

In an inelastic collision:

- **Momentum is conserved**
- **Kinetic energy is not conserved**
- Some kinetic energy is transformed into other forms (e.g. **heat, sound, deformation**)

🧠 Revision Tips

- Momentum is always conserved in collisions:

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

- Kinetic energy is conserved **only in elastic** collisions
- In inelastic collisions, use separate calculations for energy before and after
- Watch out for **signs**: negative velocities mean opposite direction